

Stuyvesant Faculty Perspectives: Mr. Peter Brooks: A Crafter Who Crafts His Own Universes in a Machine

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Mr. Peter Brooks has taught mathematics and computer science at Stuyvesant since 2005. He received his Bachelor's degree in mathematics from California Institute of Technology, an M.A. in economics from The New School, and an M.A. in Math Education from Teachers College, Columbia University. In 1984, Mr. Brooks founded MicroMind, Inc., a software development company which he ran for 13 years. Along the way, he co-developed the first interpreter for the C programming language and later created one of the first World Wide Web browsers, for which he received a national award. He also created seven other software products for the market. In 2003, he was invited to Stuyvesant to talk to students about advanced mathematics and computer science ideas. Soon after that experience, he gave up full-time software development to get his teaching license. Since then, he has taught classes such as the annual Introduction to Computer Science, Human Reasoning, and Artificial Intelligence. (Math Survey, 2015)

1 The Road That Led to Stuyvesant

Growing up with a craftsman father made Mr. Brooks interested in envisioning an idea and recreating it, which Computer Science (CS) is all about. His father, a dollmaker, built intricate visions into toys and dolls, and Brooks has crafted his own tiny universes in the mind of a machine.

This beloved teacher did not always think that he would be entering this field. When he was in tenth or eleventh grade, his aptitude and deep enthusiasm for physics and mathematics had convinced him that it would be remarkable to become a mathematical nuclear physicist. What convinced the 16-year-old Brooks even further was the way those three words sounded. However, the change of direction occurred when he was trying out various courses and found his true love: CS, obviously!

After running his own software development company and before coming to Stuyvesant, Brooks consulted on Wall Street, allowing him to leave work at 5 PM, spend time with family, work on outside projects, and most definitely read his favorite nonfiction books. He discovered that teaching consumes a great deal of time outside of the school hours, making it infeasible to work on the significant projects that he is used to creating. However, the interaction with students and watching them incorporate their artistic abilities in CS makes teaching worthwhile for him.

2 The Joy of Teaching

To Brooks, teaching the annual Introduction to Computer Science course is a golden opportunity. He can introduce an entirely new topic to students and redirect them into CS from a completely different field. Unlike in more advanced courses like Advanced Placement CS, in intro courses, he is able to “grab [our] head[s], shake it, and make it enthusiastic!” About six years ago, Brooks had created a year-long course named

Human Reasoning, which was based on critical thinking. Students were taught about fallacies, intellectual biases (identifying and getting around them), mental calculations, estimation, how to guess, and humor. Then he became interested in teaching mechanical thinking and started teaching Artificial Intelligence.

Brooks does not want students to just “learn another [EXPLETIVE] language with its own syntax and grammar,” but to be able to use it for eye-opening artistic or scientific work. He provides students with a canvas for the competition for the best personal homepage, which ends up being a “cornucopia of creativity” since some employ their graphic design skills. Grading work is always a prickly for teachers, but the needle’s tip is sharper for CS teachers. To grade a CS exam, one has to think from the perspective of a computer and decide what mistakes the computer would have detected. Doing this for over 150 students is definitely a feat, making the Stuyvesant community appreciate Brooks and the other CS teachers greatly.

According to Brooks, the CS Department has wonderful camaraderie, where the teachers observe how each other teaches, learn each other’s materials, and develop ideas together. He describes the tight bonding as “warming to the soul.” This group has many discussions and at times, arguments, on what specific materials should be taught to the students, as there are numerous intriguing subjects in CS to choose from. Unlike teachers in the other departments in Stuyvesant, the CS Department teachers have a greater scope for developing their own curriculum. This greatly benefits students as they can learn fascinating topics that many students in other schools cannot access. When teachers can choose topics they are very interested in, students naturally absorb the enthusiasm.

A common difficulty that Stuyvesant teachers face in remote learning is not being as familiar with the students as they would have been when teaching in person. Brooks falls in this category and finds the Zoom setting not very beneficial in conversing and being acquainted with all the students. In addition, Zoom fails to allow the students to capture his amusing nature and witty humor. To compensate for the limitation of learning that students obtain virtually, Brooks thoroughly looks through every homework assignment and gives feedback through his homework server instead of just scanning through them for completion as done before the pandemic.

3 Perspectives on Computer Science

CS is a field that has been expanding our horizons and will continue to do so in the future. It will be prevalent in many areas of expertise, such as data science. According to Brooks, data science is “taking the mass of raw data from a particular application, for instance, election statistics [...] and cleaning it, so to speak, or making it uniform so a machine can actually digest it.” Then one can compare the data and merge it with other pieces of data. This leads to the next step, which is machine learning; machines will look for patterns in the data or “gauge the strength of patterns that we see.”

Another area in which CS will leave a huge footprint is cybersecurity. Brooks states that it is “a very broad area that ranges from antivirus programs to cracking cellphones to defending a company against various cybersecurity attacks.” In today’s world, it is too easy to access other people’s information, making the environment perilous. However, the significant growth of cybersecurity has been encouraging.

The STEM fields are generally known for lacking gender diversity. Women account for only 27 percent in the STEM fields and make up 20 percent of CS professionals. Our school’s statistics demonstrate the same imbalance. All students take Introduction to CS as a requirement in their sophomore year, making the sex

ratio the same as any other mandatory course, around 50 percent. However, according to Brooks' data, as we reach senior-level CS courses, the percentage of female students drops significantly to about 25 percent.

When asked to explain the reason for the imbalance, he stated that both genders have slightly different personalities, "with men being more aggressive and show-offs." However, Brooks raises a bigger question: Should we see fewer women in CS as a problem? Instead of taking two semesters of CS, many students, especially female students, take a semester of a design course to fulfill their 5-Tech and 10-Tech requirements, like Architecture or Jewelry Design, that encourages their creativity. Urging female students to take CS is equivalent to saying CS is superior to other courses, which is a matter of opinion. Thus, although Brooks believes that there are not many women in CS classes, it is not a bad situation for women to be exposed to other potential interests which would be unavailable if they were on a CS track.

4 Sage Advice for Students

Using his own life as an illustration, Brooks advises students to "find [something] amongst the things that you deeply enjoy. Try many things because you really do not know what you will enjoy. Use your college years to experiment with lots of things. You are going to have less of a chance afterward."

According to Brooks, the old proverb for aspiring writers, "Kill your darlings," works for computer programmers as well. Just like writers, computer programmers have to let go of the draft that took countless hours, only to start a new one. Those of us who are prepared to go through such frustration are ready to choose CS as a career path.

In our modern materialistic world, we focus on creating something to make a lot of money. However, Brooks tells us, "You are a rare individual who has the talent and the enthusiasm for something that the society really wants [...] Try to choose amongst the vast number of possible ways to spend your talents in ways that matter to society. You have that rare choice that most people do not."